UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS08 - Network Models

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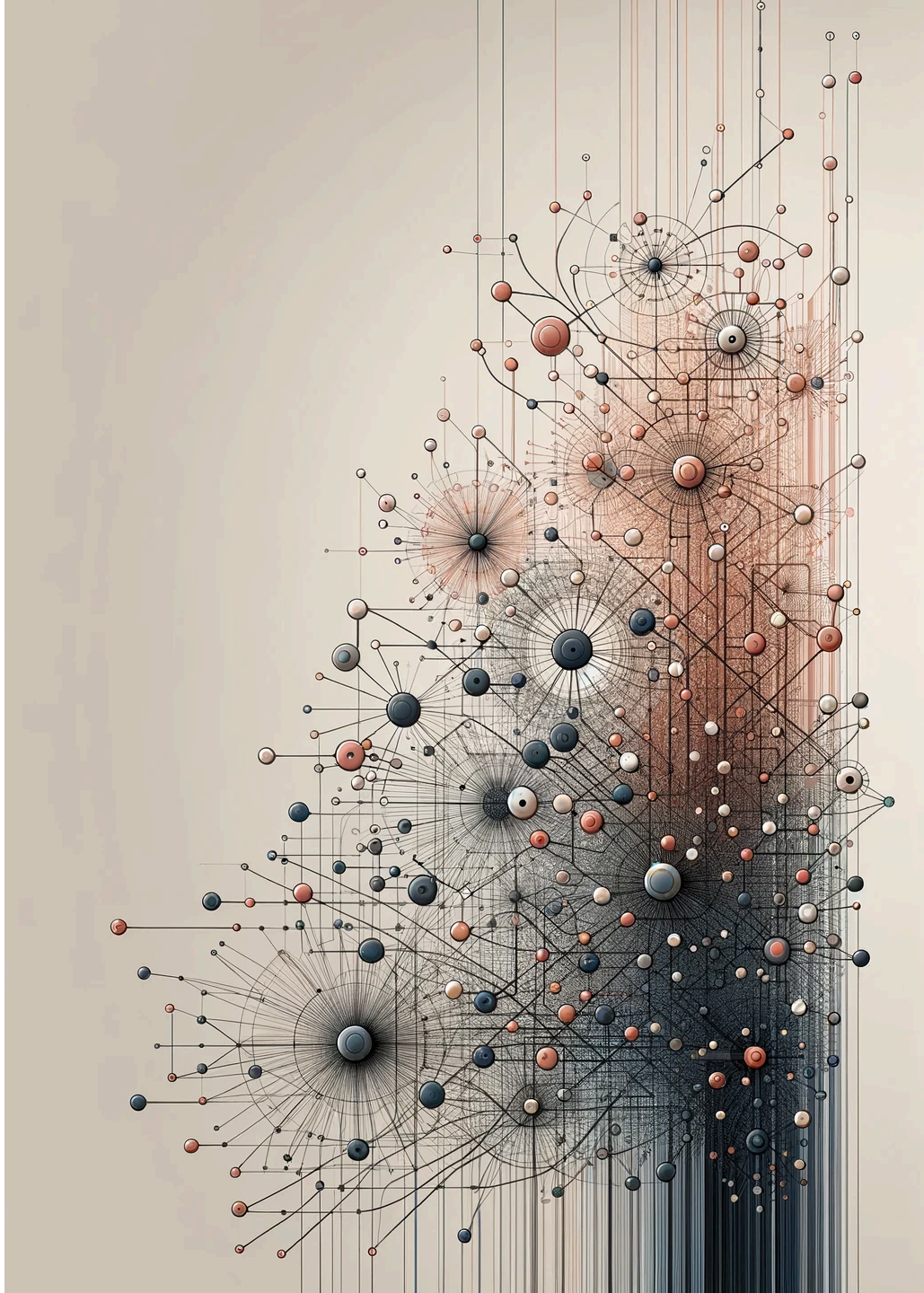


Outline

- Why do we need network models?
- Random graphs as analytical baselines
- Watts-Strogatz and the small-world mechanism
- Configuration model as a degree-preserving null model
- What these models explain, and what they do not

From observed structure to explanatory models

In `NS07` we described real networks through their **degree distributions**, **heterogeneity**, and the role of **hubs**.

Now the question changes:

- Which simple mechanisms can reproduce these structural features?
- Which features remain unexplained?
- Which models should we use as baselines or null models?

What a network model should tell us

A network model is not just a recipe to draw a graph.

It should tell us:

- **what mechanism is assumed**
- **which structural properties follow from that mechanism**
- **which properties are still missing**
- **when the model is useful as a benchmark, and when it is not**

Random graphs as a baseline

Why start from random graphs?

Random graphs are the simplest mathematical benchmark.

They help us separate three questions:

- what happens **by chance**
- what follows from a **specific structural mechanism**
- what is genuinely **non-random** in an empirical network

Gilbert's random graph model

The model is denoted by $G(n, p)$:

- n = number of nodes
- p = probability that any pair of nodes is connected

Construction: for each pair of nodes, place an edge independently with probability p.

The probability of obtaining a graph with m edges is

$$P(G) = p^m (1-p)^{\binom{n}{2} - m}$$

What the random-graph assumption means

The Erdős-Rényi / Gilbert baseline makes three strong assumptions:

- all node pairs are treated symmetrically
- edge choices are **independent**
- there is no mechanism for hubs, closure, or communities

This makes the model analytically tractable, but also structurally limited.

Connected components in random graphs

At the two extremes:

- if $p = 0$, the graph has N isolated nodes
- if $p = 1$, the graph is complete and connected

The interesting question is what happens in between.

Do connected components grow smoothly, or is there a sharp structural change?

A phase transition appears

In random graphs, the emergence of a giant component is a **phase transition**.

- for very small average degree, components remain small
- around

$$\langle k \rangle = 1$$

a giant component appears abruptly and captures a positive fraction of the nodes

This is one of the first major threshold phenomena in network science.

Why the threshold is at $\langle k \rangle = 1$

The local exploration of an Erdős-Rényi graph is approximately a branching process.

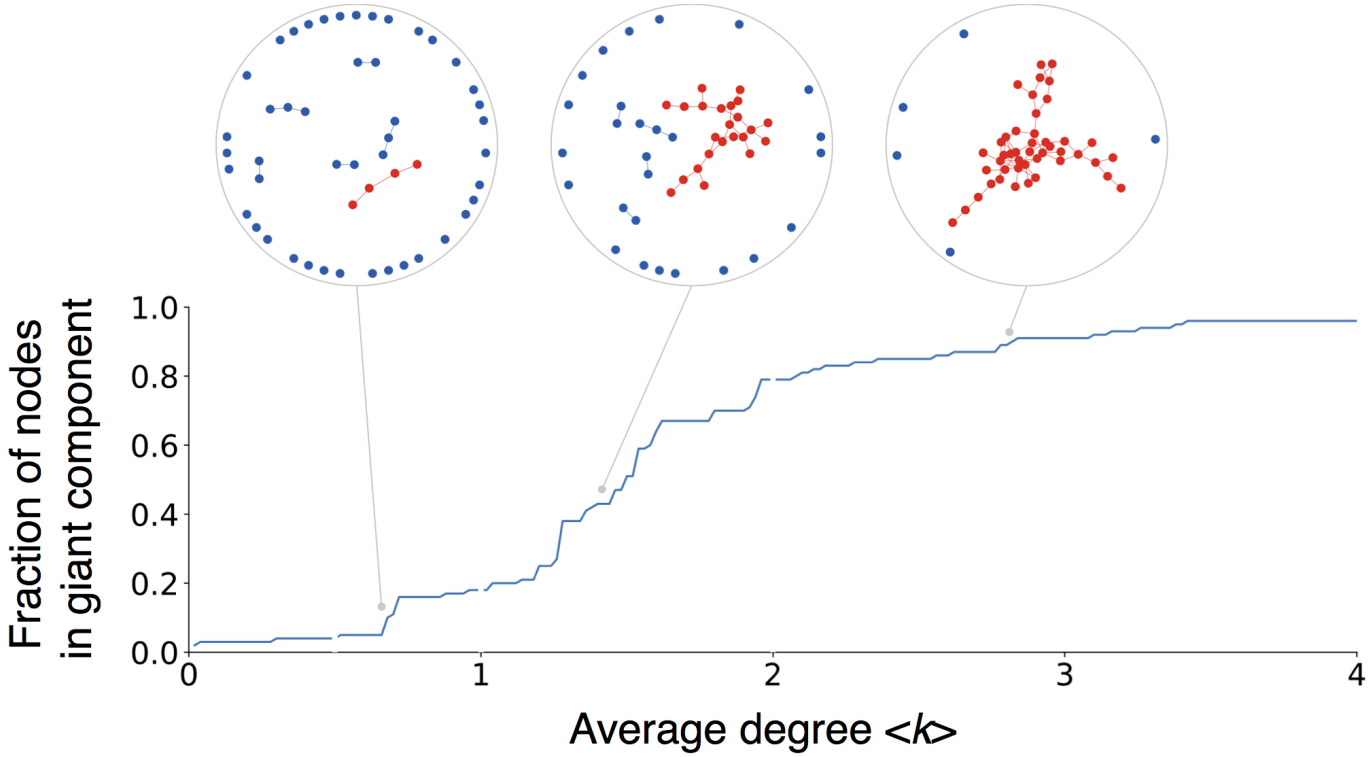
- if each reached node generates, on average, **less than one** new continuation, exploration dies out quickly
- if it generates, on average, **more than one** continuation, the explored set can keep growing

So the critical point is where the effective branching factor crosses 1:

$$\langle k \rangle \approx 1$$

Below it, components stay small. Above it, a giant component can emerge.

Giant-component transition in random graphs



Around $\langle k \rangle = 1$, the largest connected component grows very rapidly.

[Example in NetLogo]

Expected number of edges and average degree

Each node has $N-1$ possible neighbors, and each edge appears with probability p.

Therefore:

$$\langle k \rangle = p(N-1)$$

and the expected number of edges is

$$\langle L \rangle = p\frac{N(N-1)}{2}$$

Why these formulas matter

The parameter p controls the whole density regime of the model.

- increasing p increases both the expected number of edges and the expected degree
- the component transition can therefore be expressed in terms of either p or $\langle k \rangle$
- for large networks, $\langle k \rangle$ is usually the more interpretable control parameter

Degree distribution in random graphs

For a node, the number of neighbors is the number of successes in $N-1$ independent Bernoulli trials.

Hence the degree distribution is binomial:

$$P(k) = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

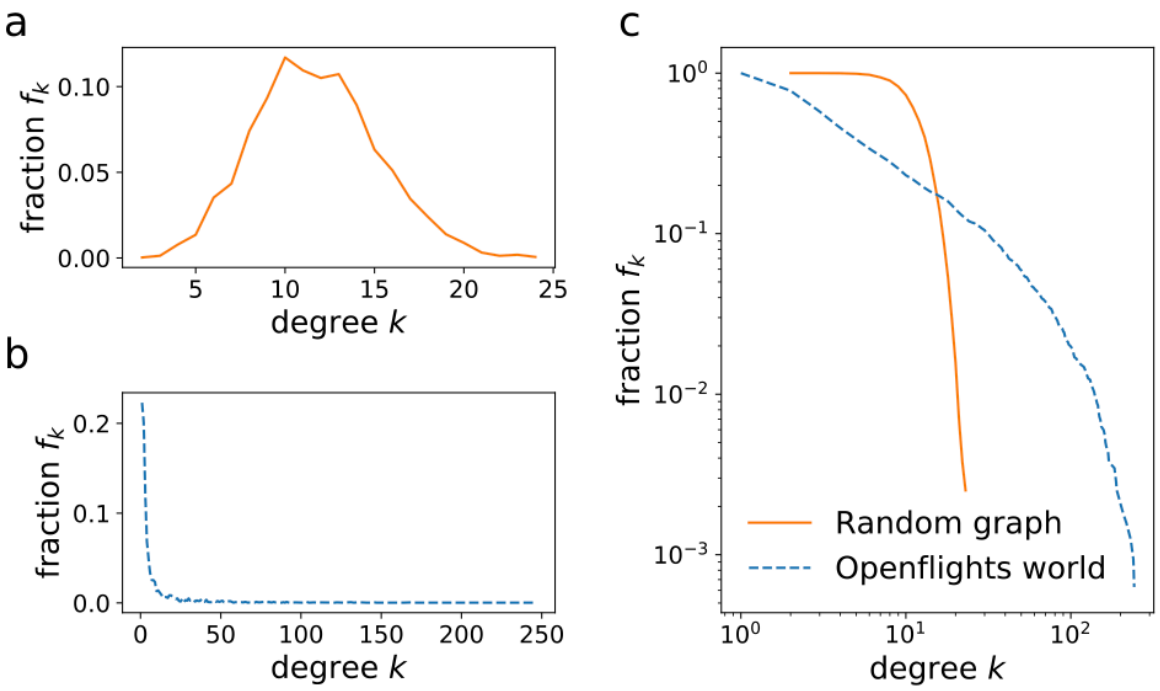
For large N and moderate density, degrees concentrate around their mean.

Narrow heterogeneity in random graphs

Because the degree distribution is narrow:

- most nodes have similar degree
- the average degree is a good summary of the distribution
- strong hubs do not emerge

This is the opposite of the broad heterogeneity seen in many real networks.



Short paths in random graphs

Assume, approximately, that each node has degree k.

Then the number of nodes reachable within distance d grows roughly as

$$N_d \sim k(k-1)^{d-1} \sim k^d$$

Solving for the number of steps needed to cover the network gives

$$d_{\max} \sim \frac{\log N}{\log k}$$

Implication: random graphs are small worlds

Random graphs already reproduce one important empirical feature:

- average path lengths grow only logarithmically with system size

So random graphs explain **short paths**, but not the other key features of many real networks.

Clustering in random graphs

The clustering coefficient of a node can be interpreted as the probability that two of its neighbors are connected.

In a random graph, edges are independent, so

$$C_i = p = \frac{\langle k \rangle}{N-1} \sim \frac{\langle k \rangle}{N}$$

For realistic large networks, this is usually very small.

What random graphs explain, and what they miss

They explain:

- short path lengths
- threshold phenomena such as giant-component emergence
- analytically tractable baselines

They miss:

- high clustering
- hubs and strong degree heterogeneity
- communities and degree-preserving structure

Random graphs as null models

Random graphs remain useful because they answer a precise question:

> What would we expect if edges were placed independently at random?

That makes them valuable as:

- **null models**
- **first analytical benchmarks**
- **reference points** for significance tests and structural comparison

From random graphs to small worlds

The problem left open by random graphs

Real networks often combine two properties:

- **short paths**
- **high clustering**

Random graphs explain the first one well, but not the second.

The next model asks whether a small number of shortcuts can preserve clustering while collapsing distances.

Watts-Strogatz and the small-world mechanism

The small-world design question

A regular lattice has:

- high clustering
- long path lengths

A random graph has:

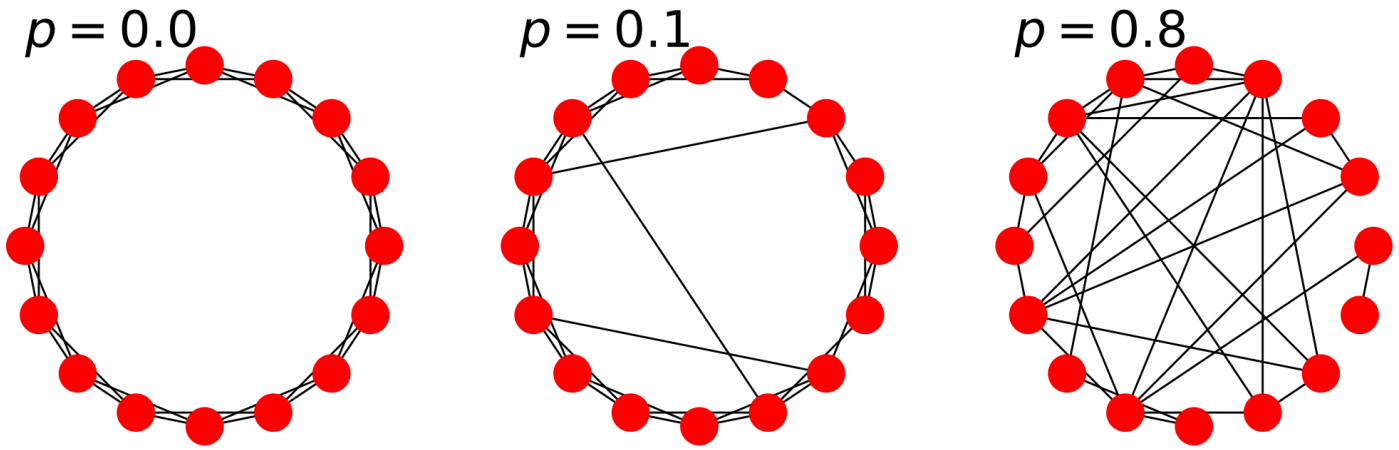
- short path lengths
- low clustering

The Watts-Strogatz model tries to interpolate between these two regimes.

The Watts-Strogatz model

1. Start from a ring lattice where each node is connected to its k nearest neighbors.
2. Rewire each edge with probability p.
3. Keep the total number of nodes and links fixed.

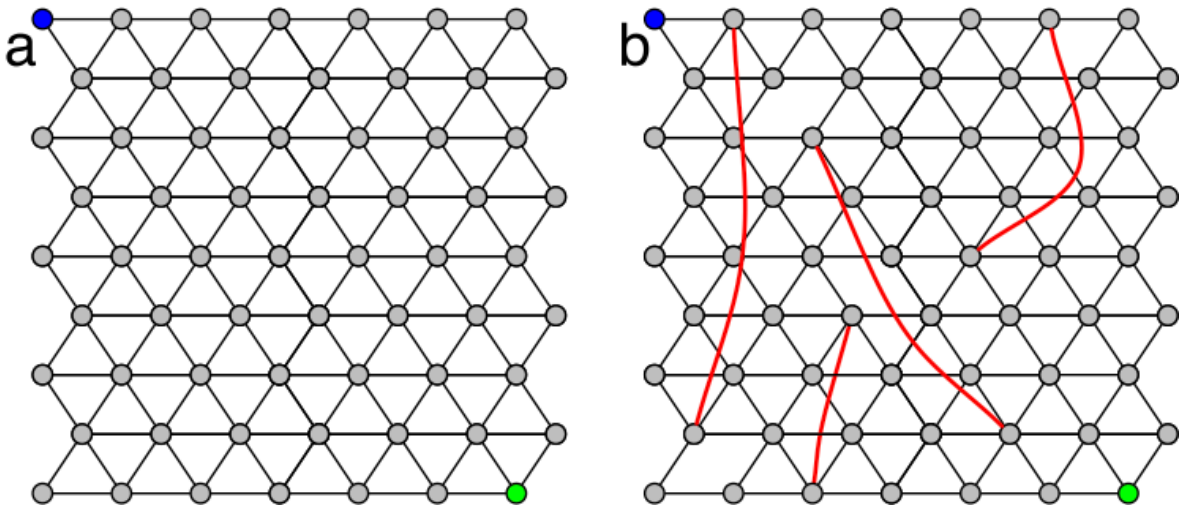
The key mechanism is the creation of a few **shortcuts** on top of a clustered local structure.



Why a few shortcuts matter so much

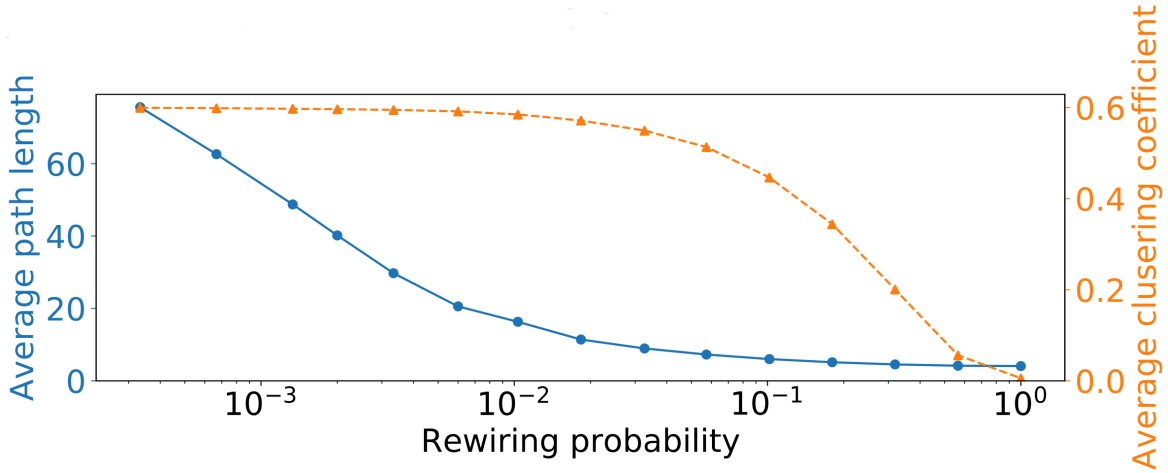
Regular lattices have long paths because movement is almost entirely local.

Even a small amount of rewiring can create long-range shortcuts that drastically reduce distances, while leaving many local triangles intact.



Effect of the rewiring probability

- $p = 0$: regular lattice
- $0 < p \ll 1$: small-world regime
- $p = 1$: random graph



The small-world regime is the narrow region where paths become short before clustering collapses.

What Watts-Strogatz explains

For intermediate rewiring probability:

- average path length becomes close to that of a random graph
- clustering remains close to that of the original lattice

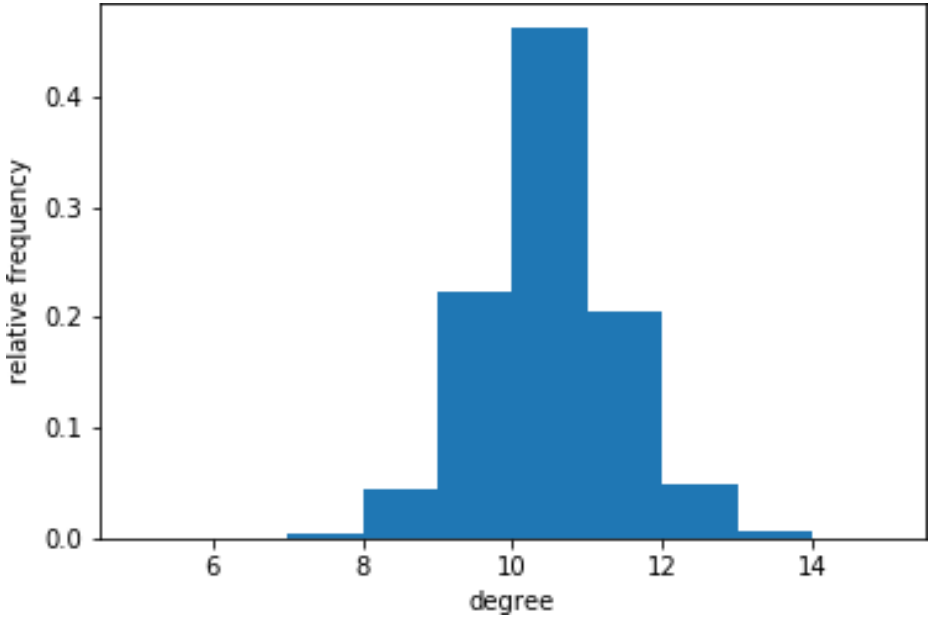
This explains the coexistence of **short paths** and **high clustering**.

What Watts-Strogatz does not explain

The degree distribution remains narrow.

So the model:

- does **not** generate hubs
- does **not** reproduce strong degree heterogeneity
- is a model of **shortcut creation**, not a model of hub emergence



Interpretation of the small-world mechanism

The Watts-Strogatz model is useful when the key question is:

> How can a network keep local cohesion and still have globally short paths?

It is less useful when the key empirical feature is a broad degree distribution.

Preserve the degree sequence

Why degree-preserving null models are needed

Suppose a real network has high clustering or visible communities.

A natural question is:

> Are these features genuine, or are they already implied by the degree sequence alone?

To answer that, we need a null model that preserves degrees while randomizing the rest.

The configuration model

The configuration model: idea

Input a degree sequence

$$(k_1, k_2, \ldots, k_N)$$

and generate a random network that preserves those target degrees.

This is not a growth model and not a social mechanism.

It is a **degree-preserving random graph model**.

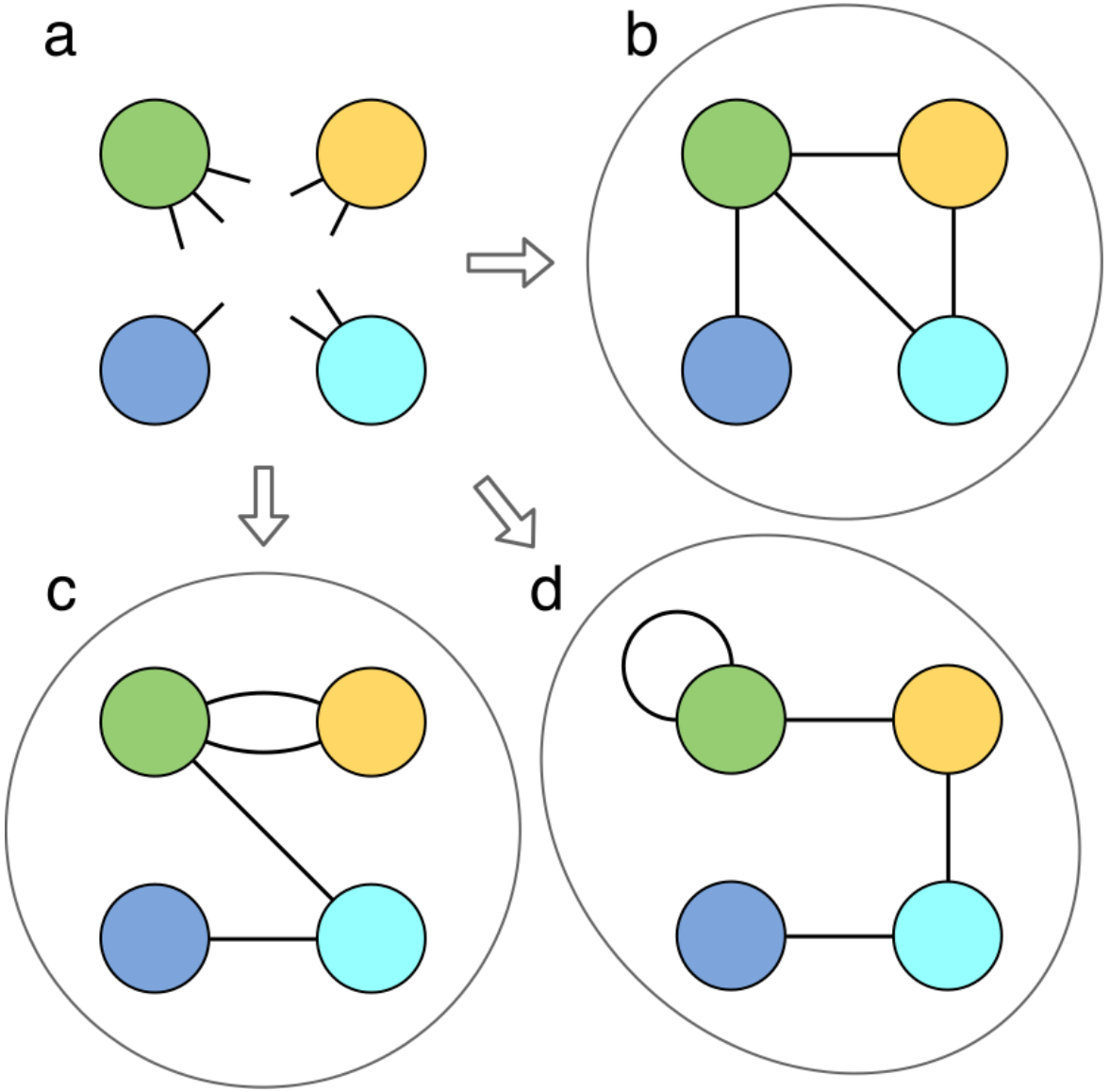
The configuration model: algorithm

1. Create N nodes.
2. Give node i exactly k_i stubs.
3. Pair stubs uniformly at random until all stubs are matched.

Necessary condition:

$$\sum_{i=1}^{N} k_i = 2m$$

so the total number of stubs must be even.


a
b
c
d

The configuration model: properties

Because the degree sequence is fixed:

- the degree distribution is preserved by construction
- path lengths are often short, as in random graphs with the same degree heterogeneity
- clustering is usually low unless extra mechanisms are added
- there is no built-in community structure or triadic closure

Degree-preserving randomization of an observed network

A closely related practical technique is **edge swapping**:

1. start from the observed graph
2. pick two edges at random
3. swap endpoints if this keeps a simple graph
4. repeat many times

This randomizes structure while preserving every node degree.

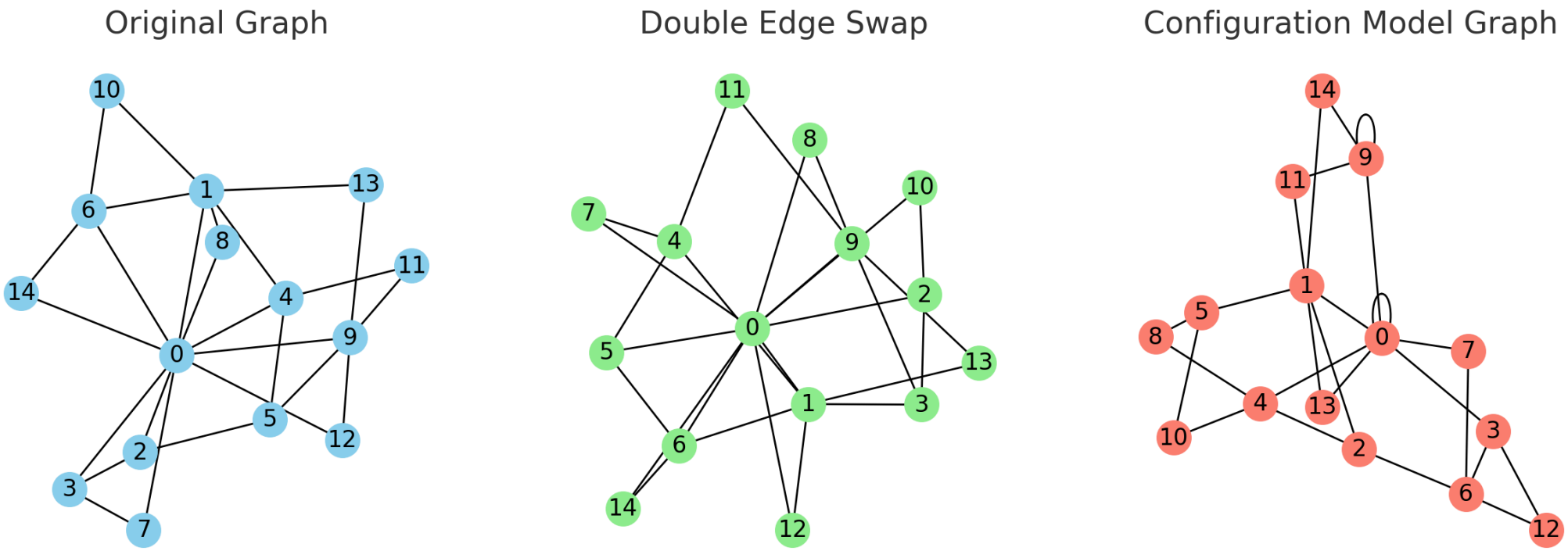
What this null model tests

The configuration model is the right control when asking whether a property is explained by degree alone.

Typical uses:

- clustering beyond what degree heterogeneity predicts
- motif significance analysis
- community structure against a degree-preserving baseline
- modularity-based null models

Degree sequence preserved, structure randomized



The degree sequence can remain fixed while mesoscale structure changes substantially.

Limitations of the configuration model

- the basic version may create self-loops and multi-edges
- not every degree sequence is realizable as a simple graph
- clustering, communities, geometry, and growth mechanisms are absent
- it explains **what degree alone implies**, not how the network formed

Comparing the three model families

Model	Key mechanism	Explains well	Misses	Typical use
Erdős-Rényi / Gilbert	independent random edges	short paths, thresholds, analytical baselines	clustering, hubs, communities	first null model
Watts-Strogatz	local lattice + shortcuts	short paths + high clustering	hubs, broad degree heterogeneity	small-world explanation
Configuration model	fixed degree sequence + random pairing	effects of degree heterogeneity	growth, closure, communities	degree-preserving null model

What remains unexplained

These models are all useful, but they leave one major empirical question open.

Many real networks contain **hubs** and **broad degree distributions**.

None of these models explains how such heterogeneity **emerges through growth**.

Next step: growth and preferential attachment

To explain the emergence of hubs we need models in which:

- the network grows over time
- new links are not placed uniformly at random
- connectivity itself changes future attachment probabilities

This leads to preferential attachment models.

Reading material

References

[ns1] **Chapter 5 (Network Models)**

Q&A